SE(3)-WAM: What Should a Robot Remember and Predict for Dynamic Manipulation?

Anonymous CVPR submission

Paper ID *****

Abstract

Dynamic manipulation requires a robot to coordinate contact with a scene that continues to change between observation and execution. This raises a question beyond predicting more of the scene: which historical and future information actually improves the action? We study history encoding, future representation, and world–action coupling in a common dynamic-manipulation framework. Our guiding hypothesis is that prediction should preserve target motion and task-relevant context at the granularity required by interaction. A completed fifteen-condition future-supervision study on fourteen simulated tasks shows why this hypothesis requires controlled testing. On 1,260 paired evaluation scenes, the highest success rate is 64.13%, compared with 62.78% without future supervision, with no clear overall advantage. Dense 2-D displacement supervision reaches 59.21% and is worse under the reported multiplicity correction. These results use privileged current object state and motion history, one training seed, and an auxiliary prediction path. With a different visual encoder and reconstruction objective, a selected trajectory-supervised configuration reaches 66.79% versus 57.86% for its matched no-future control on the 1,400-scene development pool. History and combination studies expose the conditions of representation effects. We specify the remaining backbone, online information-flow, pretraining, and external evaluations that will determine the final SE(3)-WAM instance. The present evidence supports conditional design decisions, while leaving the value of deployable online foresight to those comparisons.

1. Introduction

A moving object can leave its observed position before a robot reaches it. The same problem arises when a receptacle moves during release or an obstacle changes the available approach corridor. Dynamic manipulation therefore requires spatial accuracy at the right time. Faster inference and more responsive execution reduce observation age [31], while temporal observations help distinguish motions that look similar in a single image [9, 21]. Together, these demands motivate a practical question: *what should a robot remember and predict so that its action remains useful when interaction occurs?*

The answer depends on the role of information in the task. Target translation and rotation determine where and how to make contact. A moving occluder or obstacle matters through its effect on visibility, reachability, or collision. Static geometry can remain essential even though it has no motion to predict. This suggests organizing scene evolution by interaction needs: preserve fine target information when contact requires it, and represent surrounding changes at sufficient resolution to support the decision. Both a fixed target/context decomposition and a shared representation with task-dependent readout can implement this idea. Their relative value is an empirical question.

World–action models provide a framework for learning scene evolution alongside robot actions [34, 35]. However, a future objective changes several things at once: the supervised variables, the gradients shaping shared features, and sometimes the computation used before acting. The action may benefit from auxiliary supervision even when no forecast is generated at deployment [37]. Conversely, a geometrically accurate forecast may add no useful information to an already strong state-conditioned policy. Prediction error, action improvement, and online forecast use must therefore be measured separately.

We organize the SE(3)-WAM Study around three connected questions. **Q1** asks how a finite history should encode motion. **Q2** asks which future variables, regions, and resolutions support action. **Q3** asks how world and action computation should share parameters and exchange information. Each study matches its own reference; history, future, and combination studies use distinct recipes. Cross-comparisons revisit competitive choices before robot-data pretraining.

The completed comparisons give different outcomes across recipes. Fifteen future-supervision conditions have been trained and evaluated on a 1,260-scene complement of the development set. Sparse scene or target trajectories give

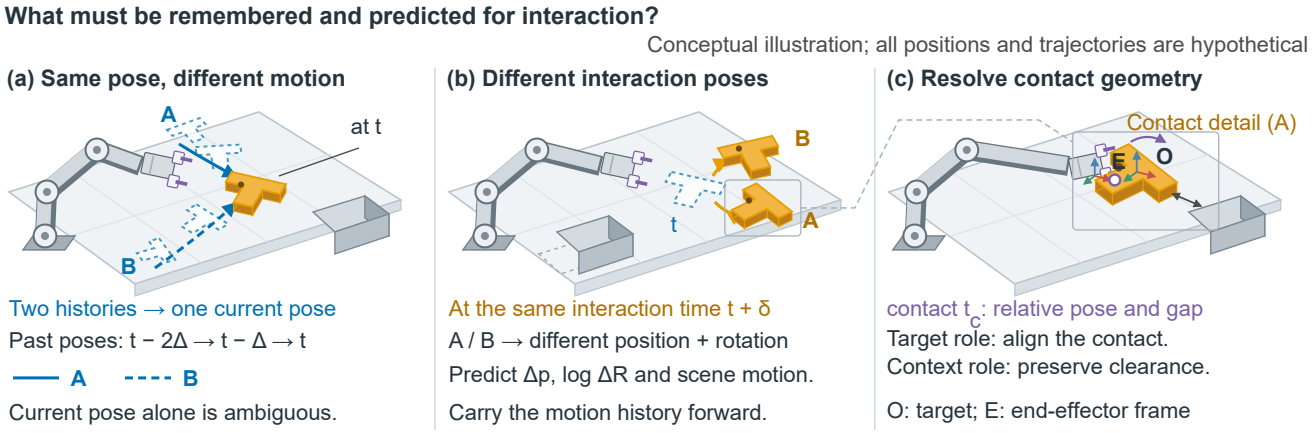


Figure 1. What information matters at interaction time? (a) Two histories meet at the same current target pose. (b) They lead to different poses at the same future time (solid A, dashed B). (c) A contact detail for A shows end-effector/target alignment and context clearance. The same asymmetric target and robot connect the panels. All trajectories are conceptual illustrations of the research question.

the highest point estimates, but neither clearly exceeds the no-future reference overall. Dense 2-D future displacement decreases success under the same comparison. A subsequent twelve-condition combination screen selects a candidate with 66.59% success on that complement. Its matched no-future control reaches 58.02%; the matched pose-and-flow control is still pending. An adapted DreamVLA reference reaches 79.60% under its own recipe. These observations motivate investigating how information is encoded and used, rather than treating the addition of future targets as an established benefit.

The current evidence supports two contributions:

- A dynamic-manipulation study protocol that separates task outcomes, available information, training supervision, and execution conditions.
- A completed comparison of fifteen future-supervision conditions, with paired results and adverse findings, complemented by history and combination studies that identify the next controlled contrasts.

The supplement details the measured configurations and the remaining design, pretraining, and transfer protocols.

2. Related Work

Dynamic manipulation and physical time. Motion-aware grasping connects prediction to reachability and interception [1]. DOMINO/PUMA couples a dynamic benchmark with historical flow and object-future supervision [9]; DynamicManip expands static demonstrations into dynamic data [16]. DynamicVLA emphasizes continuous inference and action streaming [31], while DynamicWAM conditions world–action learning on motion [21]. Temporal feature prediction [40], AHEAD's predictive latents [25], and ForeTime-VLA's future-token and event-time supervision [22] offer different ways to anticipate interaction. RTC addresses continuity under delayed chunk generation [3]; Chronos Policy explicitly represents observation and execution times [27]. These works motivate evaluating prediction together with action age, rather than identifying a simulator control interval with real-time operation.

History and object-centered information. TraceVLA encodes temporal traces as visual prompts [43]; HiF-VLA organizes past and future dynamics through motion representations [17]. PACT-WAM allocates fewer spatial tokens to older observations [19], and Long-WAM studies long context together with causal video pretraining [12]. Thus a history encoder's value depends on both temporal support and the training that makes that information usable. PointWorld explicitly assumes a static initial scene when neither prior frames nor velocities are supplied [11]. Externally moving targets make this initial-state information a central factor in Q1. DOCIR separates manipulated objects, obstacles, and the robot [8]; OA-WAM makes objects addressable within world–action modeling [20]. Our Q1 comparisons distinguish current localization, privileged motion, and visual history before comparing encodings. Role assignment is an experimental factor, not a presumed optimum.

Which future should a policy predict? SE3-Nets models scene motion through rigid transforms [4]; SE3-Pose-Nets uses structured pose dynamics for control [5]. DSR-Net accumulates a volumetric object representation over interactions [33]; PhysMani learns 3-D world representations for manipulation [38]. ATM learns point trajectories from video [29]; Im2Flow2Act uses generated object flow as a policy interface [32]. AffordTrajDP propagates contact affordances with object pose [30], and FlowWAM uses flow in

policy and world-model modes [6]. These interfaces differ in whether they describe desired motion, action-conditioned consequences, or externally moving objects. DreamVLA and DreamWAM add complementary visual, geometric, and semantic targets [36, 39]; DC-WAM instead emphasizes dynamics within RGB supervision [13]. DELE-w0.5 uses compact future latents [15]. Q2 places related target families under one action path to measure their conditional training value; its pose output does not imply a new rigid-motion parameterization.

World–action coupling and experimental attribution. Prediction enters control through different interfaces. PointWorld predicts action-conditioned 3-D point flows for candidate-action evaluation [11]; NWM simulates and ranks navigation trajectories [2]. CoT-VLA generates a visual goal before its action sequence [41], while VPP conditions actions on predictive features [10]. DreamZero jointly models world and action [34]; Fast-WAM separates training benefits from test-time imagination [37]. Our reference D uses future supervision without a forecast-to-action edge. Q3 changes this access and generation order under controlled inputs. Wang et al.'s OpenWAM supplies the Alpha foundation for our adapted baseline [28]; Yu et al.'s OpenWAM studies composable world–action programs [35]. ACT, Diffusion Policy, and $\pi_{0.5}$ supply system references [7, 23, 42]; DROID supplies the planned real-robot pretraining data [14]. Native system comparisons and matched design contrasts answer different questions and are reported separately.

3. Problem and Evaluation Protocol

Interaction in physical time. Let $\mathcal{H}_t = \{(o_i, s_i, \tau_i)\}_{i \le t}$ denote the available finite history of observations, robot state, and timestamps. A policy produces an action chunk $A_t = (a_{t,0}, \ldots, a_{t,H_a-1})$. If the latest image was captured at τ_t^s and the first command is applied after delay ℓ_t, its application times are

$$\tau_{t,j}^{\text{apply}} = \tau_t^s + \ell_t + j\Delta_a. \tag{1}$$

The relevant future is defined relative to these physical times and the task's contact event. Forecast horizons, action intervals, and computation time are different quantities. In the completed representation screen, labels use eight future samples from 50 to 400 ms, while evaluation uses synchronous policy queries and executes one action per replan. Native command delay is 50 ms for T5 and zero for the other tasks; measured computation time is not injected.

Tasks and data. The reported experiments use the migrated DB0 v0.6 data and the fourteen-task evaluation suite. The suite covers moving-target grasping, orientation-sensitive grasping, physical occlusion, moving-receptacle placement, phase-constrained passage, free rolling/sliding motion, and post-impact adaptation. The simulation infrastructure uses MuJoCo and robosuite [26, 44]. Task success is determined by each native evaluator's physical outcome; all fourteen tasks receive equal weight. The common demonstration pool contains 11,200 trajectories, 800 per task. Appendix A specifies the task and split identities; Supplementary Fig. S1 shows recorded task observations.

Information tracks. The future-supervision screen uses a current fixed-camera RGB image, exact depth/camera geometry, an eight-sample robot-state history, and privileged current target pose plus seven intervals of target kinematics. We call this the *object-motion information track*. It isolates future objectives when substantial target motion information is already available. The current-pose history study removes privileged interval kinematics and asks whether visual history supplies useful additional information. Both tracks still use privileged current target pose. A deployable visual system must estimate such quantities from allowed sensors and is evaluated separately. Current-only RGB with robot-state history is denoted H0V; it retains memory.

Evaluation partitions. The fixed140 development set contains ten scenes per task. The disjoint 1,260-scene complement contains ninety per task, giving 1,400 scenes together. The complement is independent of fixed140 membership but belongs to the same validation distribution and has now informed development. The combined 1,400-scene score includes development scenes and is used for combination selection; it is not a new out-of-distribution test. We preserve these identities in every table. Checkpoints in the completed future screen use the fixed fourth-epoch endpoint and one fresh training seed. Paired comparisons share scenes and specified rollout seeds; scenario uncertainty does not include between-training-seed variation.

Outcomes and comparison scope. For $K = 14$ tasks with n_k episodes each, the primary score is

$$\mathrm{SR}(m) = \frac{1}{K}\sum_{k=1}^{K}\frac{1}{n_k}\sum_{i=1}^{n_k} y_{mki}, \qquad y_{mki} \in \{0,1\}. \tag{2}$$

Equal per-task counts make pooled and task-balanced SR identical here. We report counts as well as percentages. Paired differences use the same episodes and the uncertainty procedure attached to the source analysis. The completed future comparison corrects its twenty-seven overall contrasts as one family. Per-task findings are exploratory.

Matched internal studies share observation fields, training exposure, action interface, and execution settings. The external-method adaptation table instead compares complete

systems with different pretraining, training budgets, and decoders. It records a useful performance reference without identifying a particular architecture's effect. Contact-time and relative-pose errors, safety outcomes, and measured action age are reserved for the mechanism and deployment studies where their measurement interfaces are available. Missing readouts are not assigned zero. Representation-specific errors retain their native units; a latent loss is not ranked numerically against a metric pose error.

4. The SE(3)-WAM Design Study

The study represents a design by (H, Z, B, G): a history encoder H, future targets Z, backbone B, and information-flow program G. Figure 2 instantiates these roles in reference D. The completed screen changes Z under a fixed reference; subsequent stages examine H, B, and G and revisit retained alternatives. A single component's point-estimate ranking does not determine the final composition.

4.1. Q1: Information Access and History Encoding

The encoder maps timestamped history into conditioning tokens, $X_t = H(\mathcal{H}_t)$. Its design includes the physical window, temporal spacing, visibility, and motion coordinates. Current RGB, stacked RGB windows, codec motion vectors, optical flow, latent history, and per-frame query compression provide different descriptions of the same recent evolution. Encoding and information access are separate factors: supplying a ground-truth velocity changes what is known, while reorganizing the same visual samples changes how it is represented.

The current-pose study holds current target pose and eight robot-state samples fixed, removes target-history kinematics, and compares eight visual-history conditions plus a freshly trained GT-history reference. The object-motion track instead supplies seven intervals of target kinematics. This separation tests the conditional value of visual history given target localization. Appendix C specifies temporal support and visibility diagnostics.

4.2. Q2: Predicting Interaction-Relevant Variables

Q2 separates *scope* (target, context, or both), *variables* (pose, motion, visual features), and *granularity* (spatial and temporal allocation). The completed comparison includes no future target, scene-flow only, target-pose only, their combination, alternative scene/target descriptions, and additional future supervision. Figure 3 gives the central contrasts; Appendix D provides all fifteen conditions. Scene flow also covers target regions, so the scene-flow-only condition is not a context-only control. Pure role separation and allocation require additional matched comparisons.

For the rigid-target condition, let (p_t, R_t) be the current target pose in the fixed metric frame. At physical offset $\delta_h = 0.05h$ seconds, $h = 1, \ldots, 8$, the supervised variables are

$$d_h = p_{t+\delta_h} - p_t, \qquad \phi_h = \mathrm{Log}(R_{t+\delta_h} R_t^\top)^\vee. \quad (3)$$

Predicted poses reconstruct as $\hat{p}_{t+\delta_h} = p_t + \hat{d}_h$ and $\hat{R}_{t+\delta_h} = \mathrm{Exp}(\hat{\phi}_h^\wedge) R_t$. Translation and left-relative rotation are parameterized separately. The pose readout pools sixteen meta tokens and predicts eight cumulative transforms. Its valid-target loss combines squared translation error with 0.1 times squared rotation geodesic error. This geometric parameterization does not impose equivariance on the network.

Scene-flow supervision describes metric cumulative point displacement. Alternatives describe dense 2-D displacement, codec block motion, sparse scene trajectories with visibility, or future video latents. Target alternatives use pooled DINO features or target trajectories. The added-supervision conditions predict depth, DINO features, dynamic-region RGB, or their combination. Their labels have different units, support, and dimensionality. Replacing scene flow with dense 2-D motion consequently tests a representation package, not dimensionality alone. Likewise, pose plus sparse tracks is not by itself a matched proof that fine-target/coarse-context allocation is optimal.

The completed study extracts future targets along demonstration trajectories and uses them as auxiliary supervision. Validity masks describe geometric support and visibility, not compatibility with a different action sequence. Future labels and validity masks remain in their supervision branches; neither is an action input. Deterministic readouts and generated variables have distinct interfaces. To compare uniform and heterogeneous allocation directly, the remaining study must hold role information, temporal support, and an applicable token/compute budget common, then vary the allocation within a compatible representation. A target mask used to construct a training label does not establish deployable target assignment.

4.3. Reference Training and Action Generation

The common Q2 reference uses a four-layer transformer of width 768 with eight attention heads. A frozen DINOv3 ViT-L/16 encoder [24] provides 196 tokens from a 224×224 current image. The D backbone shares attention projections and uses separate feed-forward networks for world/meta, flow, and action streams. Action attends to meta and action tokens; it does not attend to predicted flow or flow hidden states. Future objectives can still shape shared trainable features through their gradients.

Action generation uses flow matching [18]. With normalized action target a_1, source $a_0 \sim \mathcal{N}(0, I)$, and interpolation time $u \in [0, 1]$,

$$a_u = (1-u)a_0 + u a_1, \quad \mathcal{L}_A = \mathbb{E}\|v_\theta(a_u, u, X_t) - (a_1 - a_0)\|^2. \quad (4)$$

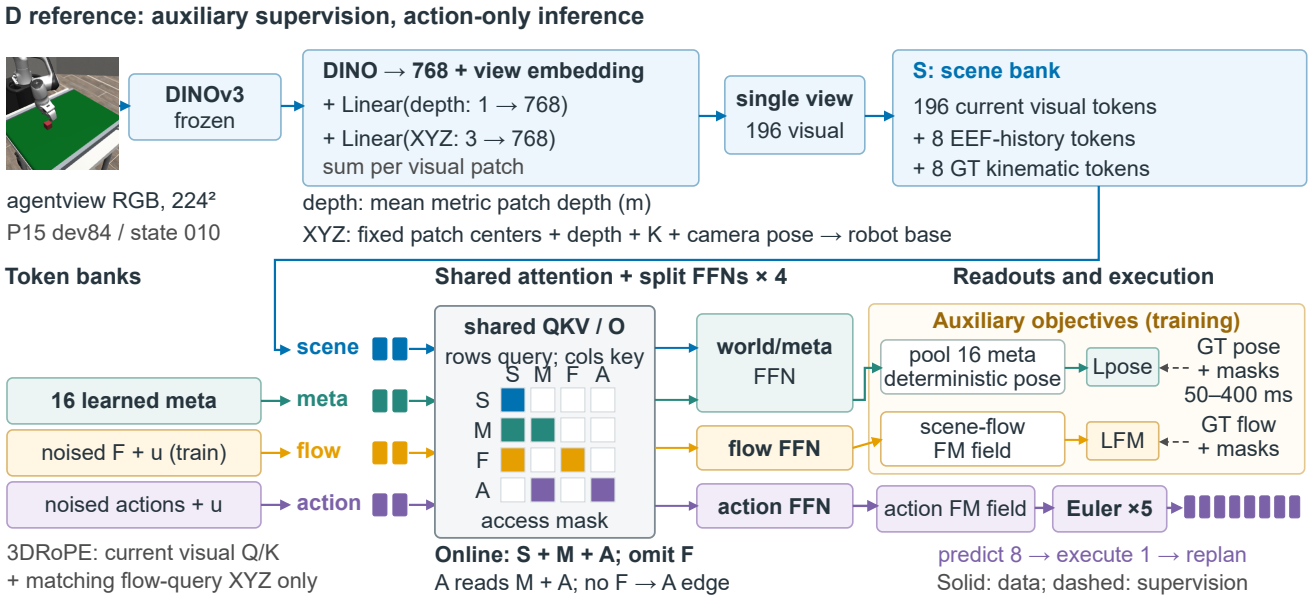


Figure 2. Implemented reference D, with P03's pose and scene-flow targets shown. RGB with depth/camera geometry, robot history, and privileged target state form the scene bank; learned meta tokens are separate inputs. The mask gives query-to-key access for scene (S), meta (M), flow (F), and action (A). Within each of four layers, the streams share attention projections and use stream-specific FFNs. Pose is a deterministic readout; scene flow and action use flow-matching fields. Future labels and masks provide training supervision (dashed). Deployment solves only action with five Euler steps, executes the first E7 command, and replans. Appendix B gives embedding and target definitions. The RGB inset is a real benchmark observation; the diagram describes D's computation.

The complete objective adds only the enabled future losses, $\mathcal{L} = \mathcal{L}_A + \sum_{r \in Z} \lambda_r \mathcal{L}_r$. Masks restrict each loss to its valid support. A missing head contributes no measurement; raw total losses with different terms are not compared as one quality metric. Appendix B specifies the reference source distributions, coordinates, and implementation settings.

The action chunk contains eight seven-dimensional commands: translation increment, left-relative rotation vector, and a gripper command. At deployment, five Euler steps generate the action chunk, its first command is executed, and the policy replans. Conditioning caching is enabled; temporal and velocity caching are disabled. The training/evaluation translation bounds are respectively 0.05/0.10 m per axis. Only the action stream is solved online in the completed Q2 screen.

4.4. Q3: Backbone and Information Flow

The next backbone comparison holds selected H/Z and the reference action-only deployment program G_0 fixed. It compares D's stream-specific feed-forward organization with an adapted E world/action parameter organization. Both are freshly trained; the proposed active-backbone parameter difference is targeted within 5%. The matched backbone results are **[pending]**.

The subsequent seven-condition information-flow study holds $H/Z/B$ fixed: R0 removes the studied future loss; R1 retains it without cross-stream edges; R2 allows future-to-action access; R3 allows action-to-future access; R4 allows both; R5 generates future then action; R6 generates action then future. R1 also compares action-only and joint deployment of the same checkpoint. These contrasts distinguish supervision, information access, and generation programs. The sequential conditions use clean ground-truth first-stage values for second-stage training and predicted values at deployment, so their difference includes teacher forcing and its deployment mismatch.

The current generative interface supports scene flow; sparse tracks or video latents require a corresponding generated-state definition. Five joint calls and five-plus-five sequential calls match per-modality steps but differ in total compute, which is measured separately. Appendix E specifies the interface, cost comparison, and online forecast interventions.

5. Experiments

We distinguish three study recipes. The future screen uses current RGB, DINO-L, GT target kinematics, Alpha off, and four-epoch cosine decay. The history screen uses current GT pose, DINO-L, and pose/flow/Alpha supervision, with one-epoch cosine decay then a constant learning rate. Combinations use full T8 RGB, GT kinematics, DINO-S or CNN, Alpha off, and four-epoch cosine decay. All retain

Table 1. System references on the 1,260-scene complement (fourteen tasks, ninety scenes each). Training seed is 0. Methods retain their adapted input, initialization, and execution pipelines; budgets and fixed140 counts are in Appendix B. Scores compare complete systems, not matched architectures. The lower block receives GT target pose and interval kinematics.

System	Initialization / study condition	Successes /1,260	SR (%)
Adapted published systems			
ACT [42]	From scratch	424	33.65
Diffusion Policy [7]	From scratch	432	34.29
$\pi_{0.5}$ [23]	Official `pi05_base`	751	59.60
DynamicVLA [31]	Adapted native pipeline	363	28.81
OpenWAM-Alpha [28]	Official Alpha foundation	708	56.19
DynamicWAM [21]	Official full initialization	670	53.17
DreamVLA [39]: 20 epochs	Independent 20-epoch training schedule	1,000	79.37
40 epochs	Independent 40-epoch training schedule	1,003	79.60
Controlled-study systems with privileged target information			
P00: no future supervision	Current image; frozen DINO-L	791	62.78
P03: scene flow + pose	Current image; frozen DINO-L	786	62.38
I: scene tracks + pose	Full visual window; CNN; target RGB reconstruction	839	66.59

robot history (Appendices B–D).

5.1. System References and Training Conditions

Table 1 places the study against adapted action and world–action models on the same fourteen-task evaluation scenes. The two DreamVLA schedules reach 1,000/1,260 and 1,003/1,260 successes. The study's combination candidate I reaches 839/1,260, and the no-future reference reaches 791/1,260. These are system-level observations. Candidate I and the reference have privileged target information; the adapted methods differ in initialization, training exposure, observation processing, and action execution. The table therefore motivates improving the current system while retaining the stronger reference, without attributing the gap to future representation.

Within the fifteen-condition future screen, all task networks start fresh with training seed 0 and run four epochs, 40,104 updates, and 641,664 sampled chunks at global batch 16. They use the fixed fourth-epoch checkpoint. Training is distributed over eight or sixteen devices with the same global batch; topology histories are retained in Appendix B. A single trained endpoint per condition supports paired scene comparisons but does not measure training-seed robustness. Current pose, motion history, action objective, and one-step receding-horizon execution are held common across this screen.

5.2. Q2: Which Future Supervision Helps?

Adding scene-flow and target-pose objectives. Figure 3 places the core contrasts in the full comparison. The no-future reference P00 succeeds in 791 scenes. Scene-flow P01, target-pose P02, and their combination P03 succeed in 782, 789, and 786 scenes, respectively. The combined objective does not show an overall improvement over either component or the reference in this setting. This result concerns auxiliary training with privileged motion input; online forecast consumption is a separate contrast.

Changing the predicted variables. Sparse scene tracks with target pose (P07) reach 807/1,260, and scene flow with target tracks (P09) reaches 808/1,260. Relative to P00, P07 gains 85 scenes and loses 69; P09 gains 91 and loses 74. Their raw two-sided exact McNemar p values are 0.227 and 0.213. The small net gains do not establish a reliable overall advantage. Non-rejection also does not establish equivalence: P12, for example, has the same total as P00 but exchanges 73 successes and 73 failures.

Dense 2-D future displacement with pose (P05) reaches 746/1,260, a decrease of 3.57 percentage points from P00 and 3.17 points from P03. The corresponding Bonferroni-adjusted p values over twenty-seven overall contrasts are 0.00844 and 0.01877. The comparison supports an adverse effect for this representation condition under the reported recipe. It does not isolate a universal 2-D versus 3-D ordering: P07 and P09 also use 2-D trajectories, with different support and target definitions.

Task variation and failure evidence. All fifteen conditions obtain zero successes on rigid-motion receptacle placement (`t2_se3`). On full constrained passage (`t3_full`), P00 and P06 obtain 3/90 and 4/90, with zero for the remaining conditions. Prediction changes have therefore not removed these shared task failures. P05's loss relative to P03 includes 27 fewer successes on rolling-sphere capture; its native `tray_contact` failures increase from 40 to 67 in that task. P09 gains on moving placement and replanning but

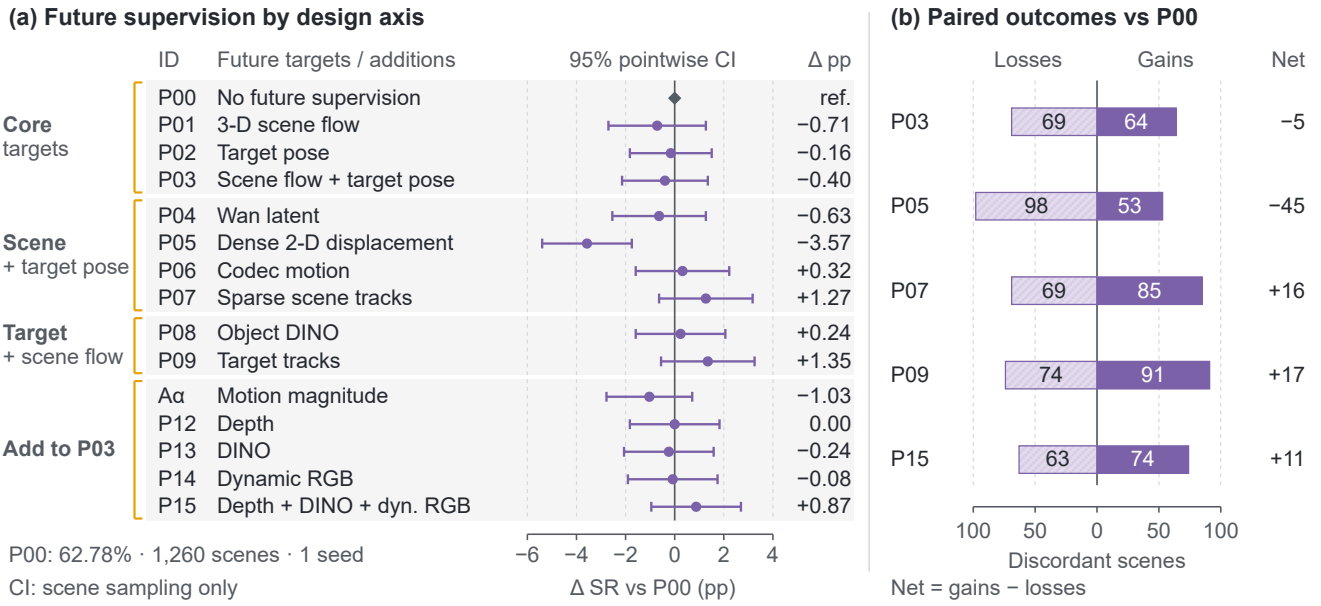


Figure 3. Future supervision at the fourth-epoch endpoint: one training seed, 1,260 paired scenes per condition (fourteen tasks, ninety each). (a) All fifteen conditions, grouped by the component changed; points show success-rate differences from P00 (62.78%), with 95% pointwise intervals from task-stratified paired bootstrap. (b) Selected contrasts expose scene-level changes: gain means P00 fails and the candidate succeeds; loss means the reverse. Net change is gain minus loss. P05 is adverse after the 27-contrast correction; P07/P09 do not establish an overall gain. Intervals condition on the trained checkpoints and exclude training-seed variability. The separate fixed140 partition is in Appendix D.

loses on orientation-sensitive grasping. These observed failure codes and counts identify where to investigate; they do not identify a representation mechanism without an aligned intervention.

Development-set ordering. P01 has the highest fixed140 point estimate, while P09 has the highest 1,260-scene point estimate. The development-set ranking does not persist across the remaining validation scenes. Appendix D reports all conditions, paired gain/loss counts, and per-task results. The evidence favors retaining competitive alternatives for joint comparison instead of treating the first ranking as a final design principle.

5.3. Combining Future Targets with Visual Encoding

The twelve-condition combination study crosses P07/P09/P15 with DINO-S or a CNN encoder and with or without current-target RGB reconstruction. It uses a full visual window and privileged target-motion information. The predefined selection score is success over all 1,400 development/evaluation scenes. Candidate I, P07 with CNN and reconstruction, obtains 935/1,400 (66.79%), including 839/1,260 (66.59%) on the complement, and is selected for matched controls (Table 2).

Its 39-scene net gain over the same P07/CNN condition without reconstruction is concentrated: phase passage gains 44 scenes and orientation-sensitive grasping gains 17, while translational placement loses 10 and XYZ grasping loses 7. Both rigid-motion placement and full passage remain at zero for all twelve conditions. Thus the selection is useful for subsequent experimentation, but its overall score does not imply uniform task improvement. The matched no-future control

Table 2. Future–encoder combinations: successes /1,400 (the development selection score, including fixed140). Rec. denotes current-target RGB reconstruction. All conditions receive a full visual window and GT target motion; DINO-S is frozen and the CNN trains from scratch. † marks selection by the development score. This condition and its matched P00 control are measured; matched P03 is pending. Dashes in the control rows denote unrun combinations. Appendix D gives task counts and paired outcomes.

	DINO-S		CNN	
Future supervision	No rec.	+rec.	No rec.	+rec.
P15: P03 + three targets	873	836	875	867
P07: scene tracks + pose	840	849	896	935†
P09: flow + target tracks	820	830	829	870
Matched controls for the selected configuration				
P00: no future	—	—	—	810
P03: flow + pose	—	—	—	Pending

Table 3. Visual history given current GT target pose and robot history. Historical target kinematics are removed. SR uses the 1,400-scene development total; Δ and its pointwise 95% scene-bootstrap interval are relative to current image, in percentage points. Appendix C gives the protocol, partition counts, and the pending GT-history reference.

Visual input	SR (%)	Δ	95% CI (pp)
Current image	53.43	0.00	—
Window	47.29	-6.14	$[-8.43, -3.79]$
Codec-MV	52.36	-1.07	$[-3.36, 1.21]$
Flow	51.43	-2.00	$[-4.43, 0.43]$
Flow12D	51.64	-1.79	$[-4.21, 0.64]$
PACT-Age	45.29	-8.14	$[-10.64, -5.64]$
Wan	50.71	-2.71	$[-5.29, -0.14]$
PACT-Uniform	47.86	-5.57	$[-8.00, -3.14]$

F, using the same CNN, visual window, and reconstruction objective, obtains 810/1,400 (57.86%), including 731/1,260 on the complement. Candidate I gains 189 paired scenes and loses 64: a net 125 scenes, or 8.93 percentage points (95% pointwise interval: 7.00–10.86), on the full selection pool. Phase passage accounts for 72 of these gains, while translational placement loses seven scenes. This comparison supports a conditional benefit of the selected future targets under this recipe; I was selected from twelve candidates on the same pool. The matched P03 control remains **[pending]**, as does confirmation across training seeds and unused scenes.

5.4. Q1: Visual History Given Current Target Pose

All eight current-pose conditions have complete scores (Table 3). Current image obtains 748/1,400 (53.43%); none of the seven visual-history encodings exceeds it in point estimate. Wan and PACT-Uniform obtain 710 and 670 successes. The fresh GT-history reference remains pending.

This experiment preserves robot-state history and current GT target pose while removing target-history kinematics. It addresses visual motion information conditional on current localization. Cross-study score differences do not estimate a history-by-future interaction. The planned matched current-pose/GT-motion and current-image/history contrasts will test whether privileged motion reduces the incremental value of visual history. Completed probe extraction alone does not answer that question.

5.5. Execution and Learning Diagnostics

Future-screen public policy calls have same-device mean times of 177.071–184.212 ms and p95 times of 178.726–185.133 ms across conditions. Peak allocated memory is 1.571–1.617 GiB. These measurements include internal perception and action computation but exclude external oracle access, I/O, rendering, loading, and priming. They come from one fixed-order profiling round; small differences do not support a stable efficiency ranking. In particular, the simulation's 50 ms action interval is not evidence of 20 Hz wall-clock deployment.

The hard-task diagnostic also identifies a supervision/execution mismatch. In the historical full-passage set, all 800 demonstrations release at the final action, with release supervision at chunk position h_7 and none at the executed h_0. This is a coverage issue under one-step deployment, rather than absence of every release label. Short diagnostic training gives inconsistent successes, and the tested first-action weighting endpoints do not establish a solution. This evidence motivates checking supervision at the executed action position before explaining difficult-task failure through future representation alone. Appendix G gives the scope and negative results.

5.6. Remaining Studies and Final Model Selection

The planned D/E and R0–R6 studies revisit retained representations and backbones before multi-seed, unused-scene confirmation (Appendix E). The selected design then undergoes recipe and scale comparisons on DROID, with a separately validated object-pose annotation pipeline. The resulting robot-data base independently initializes simulation and real-robot post-training. Single-arm DOMINO, static compatibility, and real conveyor manipulation assess transfer; measured-delay tests assess contact, action age, and end-to-end cost (Appendix F). These stages remain **[pending]**.

6. Discussion and Conclusion

Adding target and scene prediction objectives does not automatically improve a motion-informed policy: sparse trajectories give modest point-estimate gains, dense 2-D displacement is adverse under the tested recipe, and difficult tasks retain shared failures. Under the CNN/reconstruction recipe, the selected trajectory condition exceeds its matched no-future control on the development pool. Together with the history results, this makes the input and training recipe essential to interpreting representation effects.

The main screens cover fourteen simulated tasks with privileged target information, synchronous execution, and one training seed. Granularity, online foresight, design stability, and robot transfer require further tests.

References

[1] Iretiayo Akinola, Jie Xu, Shuran Song, and Peter K. Allen. Dynamic grasping with reachability and motion awareness. In *IEEE/RSJ International Conference on Intelligent Robots and Systems*, pages 9422–9429, 2021. 2

[2] Amir Bar, Gaoyue Zhou, Danny Tran, Trevor Darrell, and Yann LeCun. Navigation world models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025. 3

[3] Kevin Black, Manuel Y. Galliker, and Sergey Levine. Real-Time Execution of Action Chunking Flow Policies. In *Advances in Neural Information Processing Systems*, 2025. 2

[4] Arunkumar Byravan and Dieter Fox. SE3-Nets: Learning Rigid Body Motion using Deep Neural Networks. In *IEEE International Conference on Robotics and Automation*, 2017. 2

[5] Arunkumar Byravan, Felix Leeb, Franziska Meier, and Dieter Fox. SE3-Pose-Nets: Structured Deep Dynamics Models for Visuomotor Planning and Control. *arXiv preprint arXiv:1710.00489*, 2017. 2

[6] Yixiang Chen, Peiyan Li, Yuan Xu, Qisen Ma, Jiabing Yang, Kai Wang, Jianhua Yang, Dong An, He Guan, Gaoteng Liu, Jianlou Si, Jun Huang, Jing Liu, Nianfeng Liu, Yan Huang, and Liang Wang. FlowWAM: Optical Flow as a Unified Action Representation for World Action Models. *arXiv preprint arXiv:2607.13017*, 2026. 3

[7] Cheng Chi, Siyuan Feng, Yilun Du, Zhenjia Xu, Eric Cousineau, Benjamin C. M. Burchfiel, and Shuran Song. Diffusion policy: Visuomotor policy learning via action diffusion. In *Proceedings of Robotics: Science and Systems*, 2023. 3, 6

[8] David Emukpere, Romain Deffayet, Bingbing Wu, Romain Brégier, Michael Niemaz, Jean-Luc Meunier, Denys Proux, Jean-Michel Renders, and Seungsu Kim. Disentangled Object-Centric Image Representation for Robotic Manipulation. *arXiv preprint arXiv:2503.11565*, 2025. 2

[9] Heng Fang, Shangru Li, Shuhan Wang, Xuanyang Xi, Dingkang Liang, and Xiang Bai. Towards generalizable robotic manipulation in dynamic environments. In *European Conference on Computer Vision*, 2026. 1, 2

[10] Yucheng Hu, Yanjiang Guo, Pengchao Wang, Xiaoyu Chen, Yen-Jen Wang, Jianke Zhang, Koushil Sreenath, Chaochao Lu, and Jianyu Chen. Video Prediction Policy: A Generalist Robot Policy with Predictive Visual Representations. In *International Conference on Machine Learning*, 2025. 3

[11] Wenlong Huang, Yu-Wei Chao, Arsalan Mousavian, Ming-Yu Liu, Dieter Fox, Kaichun Mo, and Fei-Fei Li. PointWorld: Scaling 3D world models for in-the-wild robotic manipulation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026. 2, 3

[12] Wei Huang, Bohan Zhang, Chenzhi Liu, Isabella Liu, Shuai Yang, Weian Mao, Luozhou Wang, Yicheng Xiao, Weifeng Lin, Qixin Hu, Bryan Chu, Sifei Liu, Linxi Fan, Xiaojuan Qi, Song Han, and Yukang Chen. Long-WAM: Scaling the Context of World-Action Models. *arXiv preprint arXiv:2610.10528*, 2026. 2

[13] Haoyuan Ji, Lingxiang Fan, Shang Su, Yinqiao Lu, Mengkai Shi, Jun Gao, and Shuo Feng. DC-WAM: Dynamic-Centric Visual Supervision and Reasoning for World-Action Models. *arXiv preprint arXiv:2607.25918*, 2026. 3

[14] Alexander Khazatsky, Karl Pertsch, Suraj Nair, et al. DROID: A large-scale in-the-wild robot manipulation dataset. *arXiv preprint arXiv:2403.12945*, 2024. 3

[15] Fenghao Lei, Zhixiong Huang, Long Yang, Jiabao Chen, Peilin Huang, Han Fu, Zhuo Li, and Xiaoxue Ren. DELE-w0.5: Inferring Action from Future Latent State for Robotic Manipulation. *arXiv preprint arXiv:2608.22067*, 2026. 3

[16] Haoran Liao, Pengyue Wang, Shuoyu Chen, Kehan Cheng, Xuhang Chen, Yuhao Lin, Mu Lin, Zhizhao Liang, Xiaoyi Fan, Chengyi Xing, Dan Niu, Yi-Lin Wei, and Wei-Shi Zheng. DynamicManip: Enabling Dynamic Manipulation from a Single Static Demonstration. *arXiv preprint arXiv:2608.01452*, 2026. 2

[17] Minghui Lin, Pengxiang Ding, Shu Wang, Zifeng Zhuang, Yang Liu, Xinyang Tong, Wenxuan Song, Shangke Lyu, Siteng Huang, and Donglin Wang. HiF-VLA: Hindsight, Insight and Foresight through Motion Representation for Vision-Language-Action Models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026. 2

[18] Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matthew Le. Flow matching for generative modeling. In *International Conference on Learning Representations*, 2023. 4

[19] Yushan Liu, Jingjing Fan, Shoujie Li, Yifan Xie, Xiao-Ping Zhang, and Wenbo Ding. PACT-WAM: Predicting Actions and Visual Foresight with Compact Temporal Encoding for Robot Manipulation. *arXiv preprint arXiv:2602.15882*, 2026. 2

[20] Yushan Liu, Peibo Sun, Shoujie Li, Yifan Xie, Lingfeng Zhang, Xintao Chao, Shiyuan Dong, Fang Chen, Xiao-Ping Zhang, and Wenbo Ding. OA-WAM: Object-addressable world action model for robust robot manipulation. *arXiv preprint arXiv:2605.06481*, 2026. 2

[21] Yunfan Lou, Hewen Gao, Xiyu Zhu, Zhuoran Qiao, Xuan Han, Yifan Yang, Yifan Ye, Boxian Yao, and Zhibo Pang. DynamicWAM: Dual-path motion conditioning for world-action models in dynamic manipulation. *arXiv preprint arXiv:2608.00793*, 2026. 1, 2, 6

[22] Siyuan Ma, Yutian Zhang, Boshi Zhang, Qinglian Wu, Jiaqi Zhai, Dong Wei, and Xiaojin Huang. ForeTime-VLA: Causal Future-Token Distillation from a World Action Model for Conveyor-Belt Manipulation. *arXiv preprint arXiv:2608.20735*, 2026. 2

[23] Physical Intelligence, Kevin Black, Noah Brown, James Darpinian, et al. $\pi_{0.5}$: a vision-language-action model with open-world generalization. *arXiv preprint arXiv:2504.16054*, 2025. 3, 6

[24] Oriane Siméoni, Huy V. Vo, Maximilian Seitzer, Federico Baldassarre, Maxime Oquab, Cijo Jose, Vasil Khalidov, Marc Szafraniec, Seungeun Yi, Michaël Ramamonjisoa, et al. DINOv3. *arXiv preprint arXiv:2508.10104*, 2025. 4

[25] Shahram Najam Syed, Arthur Jakobsson, Haoran Hao, and Jeffrey Ichnowski. Intercepting the future: Latent-space predictive world model for dynamic VLA manipulation. In *Conference on Robot Learning*, 2026. 2

[26] Emanuel Todorov, Tom Erez, and Yuval Tassa. MuJoCo: A physics engine for model-based control. In *IEEE/RSJ International Conference on Intelligent Robots and Systems*, pages 5026–5033, 2012. 3

[27] Changhao Wang, Jaden Clark, and Shuran Song. Chronos Policy: Learning Robot Policies on Physical Time. *arXiv preprint arXiv:2610.13929*, 2026. 2

[28] Yuran Wang, Siqiao Huang, Mingleyang Li, Chenhao Zhang, Jiaqi Liang, Weiyang Jin, Yue Chen, Xuemin Chi, Donghao Zhou, Qize Yu, Yu-Kai Wang, Yuhan Rui, Shenzhe Yao, Zhen Yuan, Zhenhao Shen, Kefei Zhu, Zijie Zhu, Ning Gao, Xiaowei Chi, Guanqi He, Shanghang Zhang, Hao Dong, Lin Shao, and Hang Zhao. OpenWAM: An Open, Modular Exploration Towards Systematic World-Action Model Pretraining. *arXiv preprint arXiv:2609.07398*, 2026. 3, 6

[29] Chuan Wen, Xingyu Lin, John So, Kai Chen, Qi Dou, Yang Gao, and Pieter Abbeel. Any-point Trajectory Modeling for Policy Learning. *arXiv preprint arXiv:2401.00025*, 2024. 2

[30] Gaoyuan Wu, Ziyu Shan, Haoyang Du, Yuyao Jiang, and Ziwei Wang. AffordTrajDP: Dynamic Affordance-Guided Visuomotor Policy Learning for Robotic Manipulation. *arXiv preprint arXiv:2608.01603*, 2026. 2

[31] Haozhe Xie, Beichen Wen, Jiarui Zheng, Zhaoxi Chen, Fangzhou Hong, Haiwen Diao, and Ziwei Liu. DynamicVLA: A vision-language-action model for dynamic object manipulation. In *Advances in Neural Information Processing Systems*, 2026. 1, 2, 6

[32] Mengda Xu, Zhenjia Xu, Yinghao Xu, Cheng Chi, Gordon Wetzstein, Manuela Veloso, and Shuran Song. Flow as the Cross-Domain Manipulation Interface. In *Conference on Robot Learning*, 2024. 2

[33] Zhenjia Xu, Zhanpeng He, Jiajun Wu, and Shuran Song. Learning 3D Dynamic Scene Representations for Robot Manipulation. In *Conference on Robot Learning*, 2020. 2

[34] Seonghyeon Ye, Yunhao Ge, Kaiyuan Zheng, Shenyuan Gao, Sihyun Yu, George Kurian, Suneel Indupuru, et al. World action models are zero-shot policies. *arXiv preprint arXiv:2602.15922*, 2026. 1, 3

[35] Heng Yu, David D. Yuan, Juze Zhang, Changan Chen, Yao Feng, Michelle Baldonado, Steve Cousins, Fei-Fei Li, Jiajun Wu, and Ehsan Adeli. OpenWAM: An open framework for composable world-action models. *arXiv preprint arXiv:2610.07922*, 2026. 1, 3

[36] Shanglin Yuan, Weiheng Zhao, Xin Shi, Haoyi Jiang, Xianda Guo, Liu Liu, Wenyu Liu, Wei Sui, and Xinggang Wang. DreamWAM: Beyond RGB Future Prediction for World Action Models. *arXiv preprint arXiv:2608.04996*, 2026. 3

[37] Tianyuan Yuan, Zibin Dong, Yicheng Liu, and Hang Zhao. Fast-WAM: Do world action models need test-time future imagination? *arXiv preprint arXiv:2603.16666*, 2026. 1, 3

[38] Peng Yun, Shouwang Huang, Hao Li, Jinxi Li, Jianan Wang, and Bo Yang. PhysMani: Physics-principled 3D world model for dynamic object manipulation. In *European Conference on Computer Vision*, 2026. 2

[39] Wenyao Zhang, Hongsi Liu, Zekun Qi, Yunnan Wang, XinQiang Yu, Jiazhao Zhang, Runpei Dong, Jiawei He, He Wang, Zhizheng Zhang, Li Yi, Wenjun Zeng, and Xin Jin. DreamVLA: A vision-language-action model dreamed with comprehensive world knowledge. In *Advances in Neural Information Processing Systems*, pages 24195–24228, 2025. 3, 6

[40] Yifan Zhang, Ruiping Wang, and Xilin Chen. Dynamic behavior cloning with temporal feature prediction: Enhancing robotic arm manipulation in moving object tasks. *IEEE Robotics and Automation Letters*, 10(6):5209–5216, 2025. 2

[41] Qingqing Zhao, Yao Lu, Moo Jin Kim, Zipeng Fu, Zhuoyang Zhang, Yecheng Wu, Zhaoshuo Li, Qianli Ma, Song Han, Chelsea Finn, Ankur Handa, Ming-Yu Liu, Donglai Xiang, Gordon Wetzstein, and Tsung-Yi Lin. CoT-VLA: Visual chain-of-thought reasoning for vision-language-action models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025. 3

[42] Tony Z. Zhao, Vikash Kumar, Sergey Levine, and Chelsea Finn. Learning fine-grained bimanual manipulation with low-cost hardware. *arXiv preprint arXiv:2304.13705*, 2023. 3, 6

[43] Ruijie Zheng, Yongyuan Liang, Shuaiyi Huang, Jianfeng Gao, Hal Daumé III, Andrey Kolobov, Furong Huang, and Jianwei Yang. TraceVLA: Visual Trace Prompting Enhances Spatial-Temporal Awareness for Generalist Robotic Policies. *arXiv preprint arXiv:2412.10345*, 2024. 2

[44] Yuke Zhu, Josiah Wong, Ajay Mandlekar, Roberto Martín-Martín, Abhishek Joshi, Kevin Lin, Abhiram Maddukuri, Soroush Nasiriany, and Yifeng Zhu. robosuite: A modular simulation framework and benchmark for robot learning. *arXiv preprint arXiv:2009.12293*, 2020. 3